

VIT-AP
UNIVERSITY

DIGITAL LOGIC DESIGN

(Course Code: ECE 1003)

Module-1:Lecture-2

Number Systems

CONTENTS

Module-1 (Part-1)

- ❖ **Number systems and conversions**
- ❖ r 's and $(r-1)$'s compliment
- ❖ Binary signed and unsigned numbers
- ❖ Binary arithmetic operations
- ❖ Weighted and non-weighted binary codes

CONTENTS

Lecture-2

- ❖ Conversion of different number systems

BOOKS

Textbooks

1. M.Morris Mano, Michael D Ciletti, **Digital Design**, 5th edition, Pearson Publishers, 2013.
2. R.P. Jain, “**Modern Digital Electronics**”, 4th edition, TMH.

References

1. M.Morris Mano, Charles R. Kime, Tom Martin, **Logic and Computer Design Fundamentals**, 4th edition, Pearson Publishers.
2. C. H. Roth and L. L. Kinney, **Fundamentals of Logic Design**, 5th edition, Cengage Publishers.

Number Systems

WORKING OF AN ODOMETER

0	0	0	0	0
0	0	0	0	1
0	0	0	0	2
0	0	0	0	3
0	0	0	0	4
0	0	0	0	5
0	0	0	0	6
0	0	0	0	7
0	0	0	0	8
0	0	0	0	9
0	0	0	1	0

Decimal system
(we are used to)

Octal system

0	0	0	0	0
0	0	0	0	1
0	0	0	0	2
0	0	0	0	3
0	0	0	0	4
0	0	0	0	5
0	0	0	0	6
0	0	0	0	7
0	0	0	1	0
0	0	0	1	1
0	0	0	1	2

Binary system

0	0	0	0	0
0	0	0	0	1
0	0	0	1	0
0	0	0	1	1
0	0	1	0	0
0	0	1	0	1
0	0	1	1	0
0	0	1	1	1
0	1	0	0	0
0	1	0	0	1
0	1	0	1	0



NUMBER SYSTEMS

Decimal	Binary	Octal	Hexadecimal
0	0000	0	0
1	0001	1	1
2	0010	2	2
3	0011	3	3
4	0100	4	4
5	0101	5	5
6	0110	6	6
7	0111	7	7

Note:

$$8 = 2^3,$$

Hence each
octal number
correspond to
3 binary bits

NUMBER SYSTEMS

Decimal	Binary	Octal	Hexadecimal
8	1000	10	8
9	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

Note:

$16 = 2^4$,
Hence each
**hexadecimal
number**
correspond to
4 binary bits

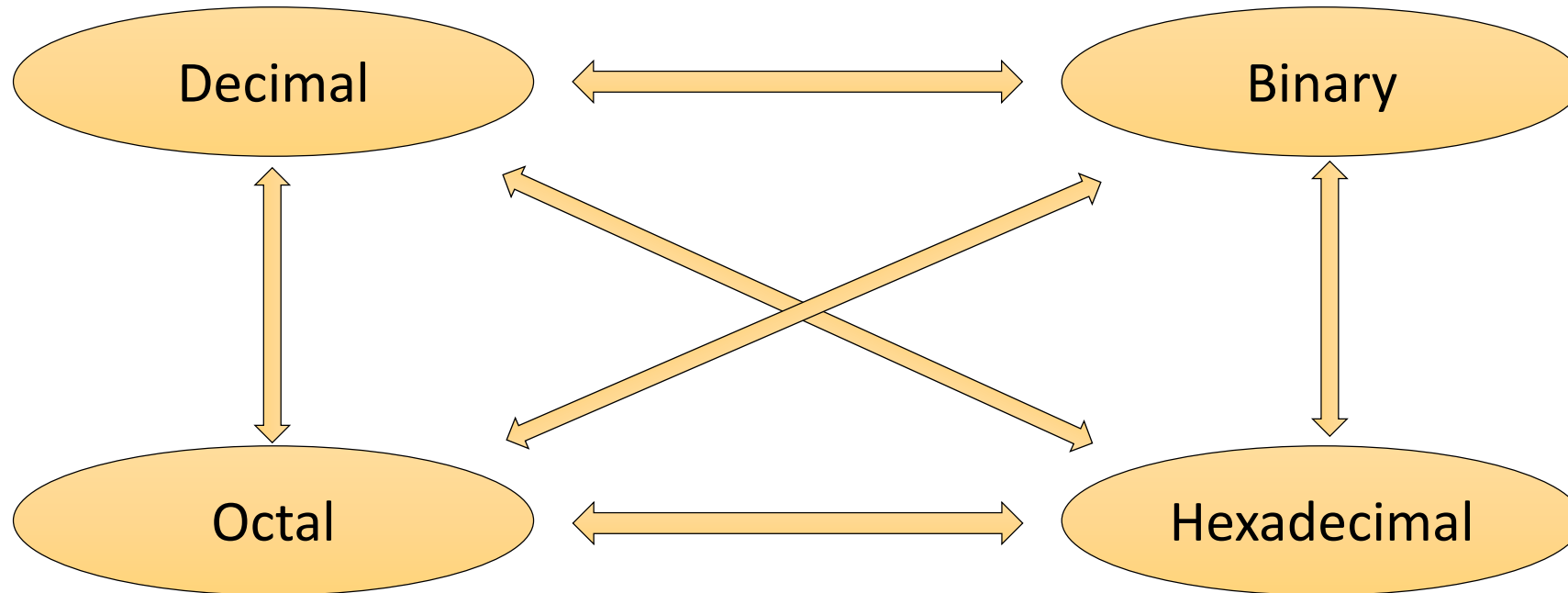
NUMBER SYSTEMS

Decimal	Binary	Octal	Hexadecimal
16	10000	20	10
17	10001	21	11
18	10010	22	12
19	10011	23	13
20	10100	24	14
21	10101	25	15
22	10110	26	16
23	10111	27	17

Etc.

CONVERSION BETWEEN NUMBER BASES

✓ The possibilities:



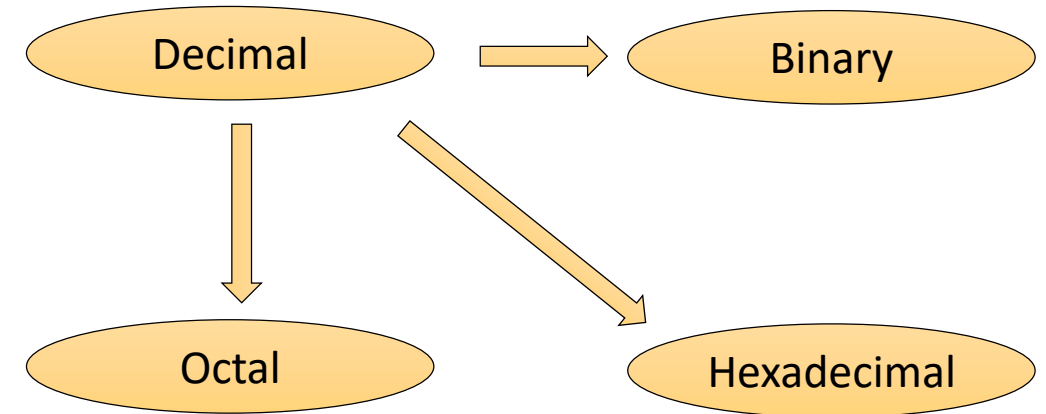
CONVERSION FROM DECIMAL TO ANOTHER BASE

Convert an Integer from Decimal to Another Base

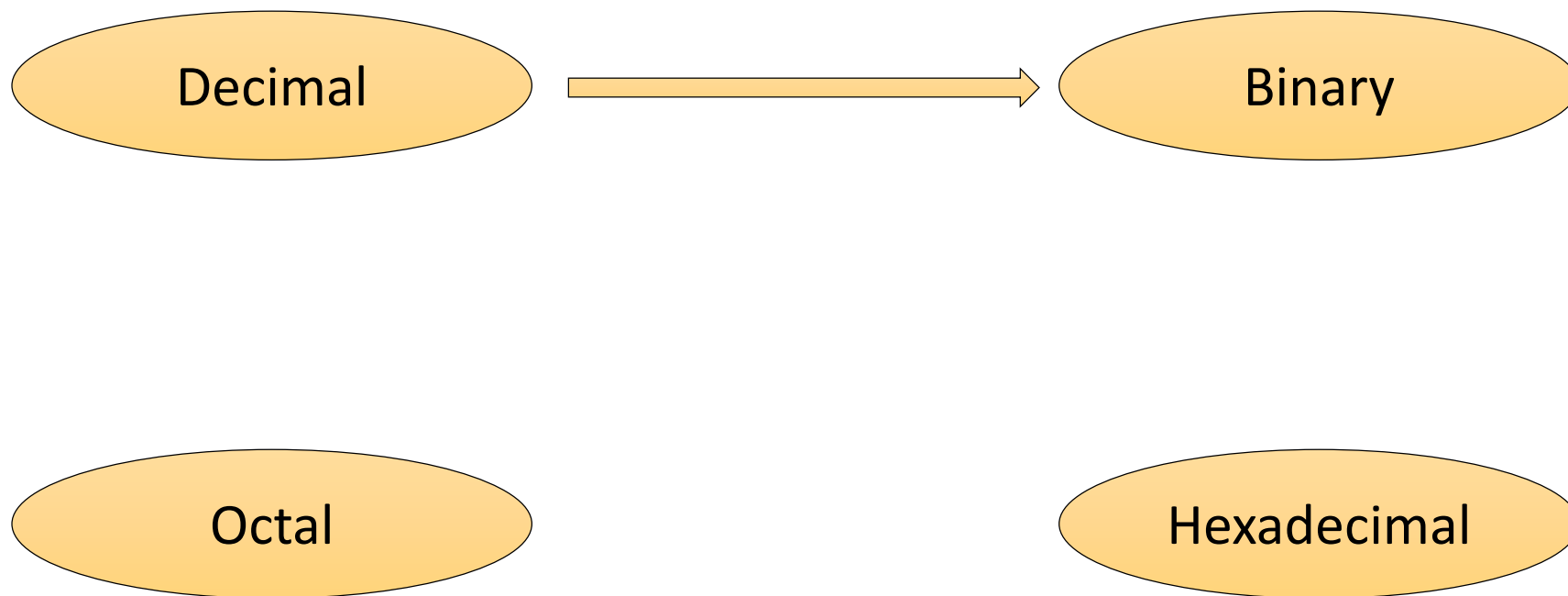
- ✓ For each digit position:
 1. Divide decimal number by the base (e.g. 2)
 2. The *remainder* is the lowest-order digit
 3. Repeat first two steps until no *divisor* remains.

Convert a fraction from Decimal to Another Base

- ✓ For each digit position:
 1. Multiply decimal number by the base (e.g. 2)
 2. The *integer* is the highest-order digit
 3. Repeat first two steps until fraction becomes zero.



CONVERSION FROM DECIMAL TO BINARY



CONVERSION FROM DECIMAL TO BINARY

Convert an Integer from Decimal to Binary

✓ Example for $(13)_{10}$:

		Integer Quotient		Remainder	Coefficient
$13/2$	$=$	6	+	1	$a_0 = 1$
$6/2$	$=$	3	+	0	$a_1 = 0$
$3/2$	$=$	1	+	1	$a_2 = 1$
$1/2$	$=$	0	+	1	$a_3 = 1$



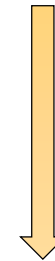
$$(13)_{10} = (a_3 a_2 a_1 a_0)_2 = (1101)_2$$

CONVERSION FROM DECIMAL TO BINARY

Convert a fraction from Decimal to Binary

✓ Example for $(0.625)_{10}$:

	Integer	Fraction	Coefficient
$0.625 \times 2 =$	1	+ 0.25	$a_{-1} = 1$
$0.250 \times 2 =$	0	+ 0.50	$a_{-2} = 0$
$0.500 \times 2 =$	1	+ 0	$a_{-3} = 1$



$$(0.625)_{10} = (0.a_{-1} a_{-2} a_{-3})_2 = (0.101)_2$$

CONVERSION FROM DECIMAL TO BINARY

✓ **Example:** Convert $(23.6)_{10}$ to binary :

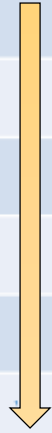
No.	÷	Quot.	Remainder
23	2	11	1
11	2	5	1
5	2	2	1
2	2	1	0
1	2	0	1



Conversion of 23

Integer part

No.	X	Carry	Fraction
0.6	2	1	0.2
0.2	2	0	0.4
0.4	2	0	0.8
0.8	2	1	0.6
0.6	2	1	0.2
0.2	...	...	...



Conversion of 0.6

Fractional part

$$(23.6)_{10} = (10111.10011)_2$$

(fractional part truncated at 5 bits)

CONVERSION FROM DECIMAL TO BINARY

✓ Example: $125_{10} = ?_2$

2	125	
2	62	1
2	31	0
2	15	1
2	7	1
2	3	1
2	1	1
	0	1

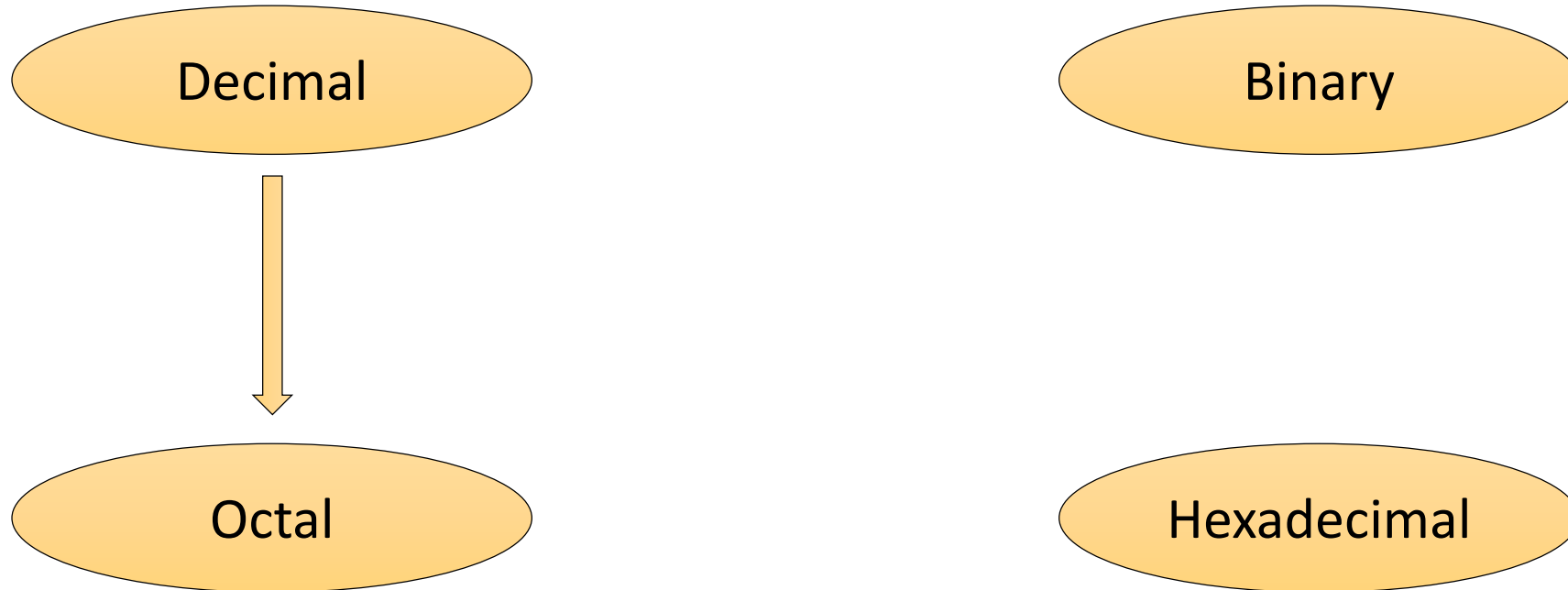


Trick:

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128	64	32	16	8	4	2	1
0	1	1	1	1	1	0	1

$$(125)_{10} = (1111101)_2$$

CONVERSION FROM DECIMAL TO OCTAL



CONVERSION FROM DECIMAL TO OCTAL

Convert an Integer from Decimal to Octal

✓ Example for $(175)_{10}$:

	Integer				
	Quotient		Remainder		Coefficient
$175/8 =$	21	+	7		$a_0 = 7$
$21/8 =$	2	+	5		$a_1 = 5$
$2/8 =$	0	+	2		$a_2 = 2$



$$(175)_{10} = (a_2 a_1 a_0)_8 = (257)_8$$

CONVERSION FROM DECIMAL TO OCTAL

Convert a fraction from Decimal to Octal

✓ Example for $(0.3125)_{10}$:

	Integer		Fraction		Coefficient
0.3125×8	= 2	+	0.5		$a_{-1} = 2$
0.5000×8	= 4	+	0		$a_{-2} = 4$



$$(0.3125)_{10} = (0.24)_8$$

CONVERSION FROM DECIMAL TO OCTAL

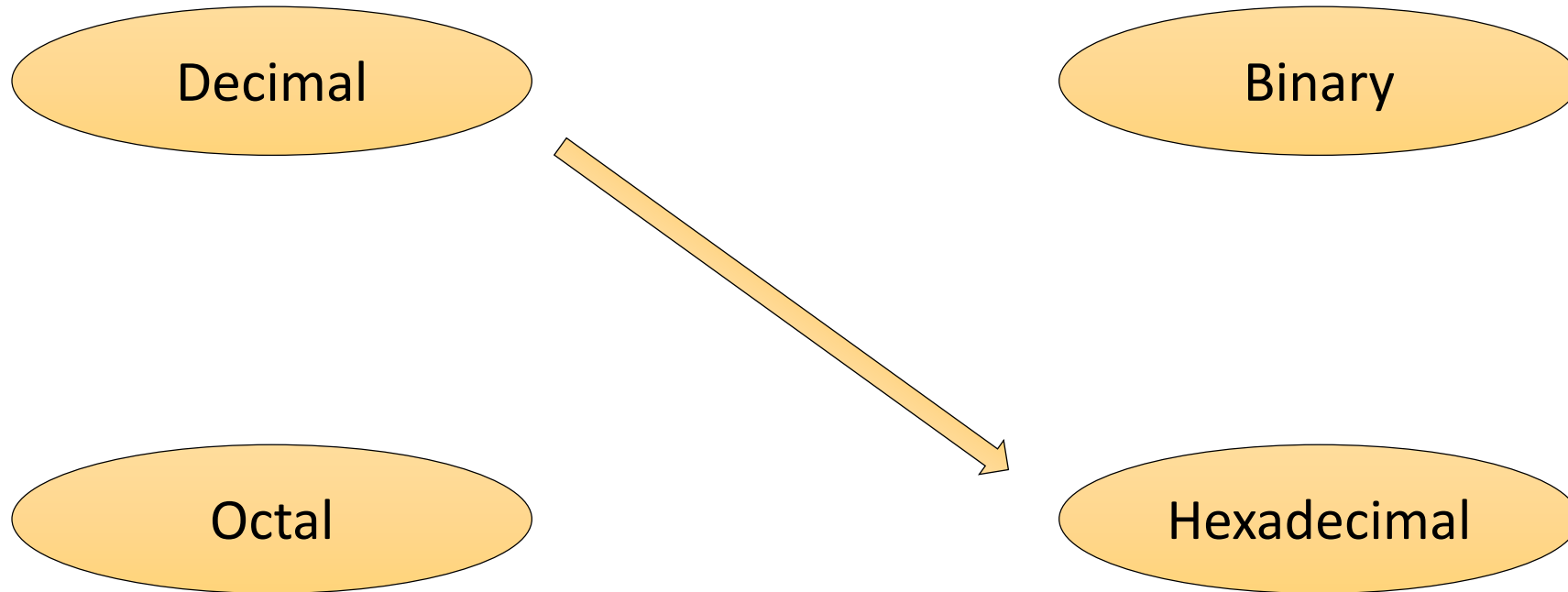
✓ **Example:** $1234_{10} = ?_8$

8	1234	
8	154	2
8	19	2
8	2	3
	0	2



$$(1234)_{10} = (2322)_8$$

CONVERSION FROM DECIMAL TO HEXADECIMAL



CONVERSION FROM DECIMAL TO HEXADECIMAL

Convert an Integer from Decimal to Hexadecimal

✓ Example for $(257)_{10}$:

	Integer				
	Quotient		Remainder		Coefficient
$257/16 =$	16	+	1		$a_0 = 1$
$16/16 =$	1	+	0		$a_1 = 0$
$1/16 =$	0	+	1		$a_2 = 1$



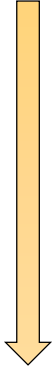
$$(257)_{10} = (a_2 a_1 a_0)_{16} = (101)_{16}$$

CONVERSION FROM DECIMAL TO HEXADECIMAL

Convert a fraction from Decimal to Hexadecimal

✓ Example for $(0.1275)_{10}$:

	Integer		Fraction	Coefficient
$0.1275 \times 16 =$	2	+	0.04	$a_{-1} = 2$
$0.04 \times 16 =$	0	+	0.64	$a_{-2} = 0$
$0.64 \times 16 =$	10	+	0.24	$a_{-3} = A$
-----	--		---	----



$$(0.1275)_{10} = (0.20A)_{16}$$

CONVERSION FROM DECIMAL TO HEXADECIMAL

✓ **Example:** $1234_{10} = ?_{16}$

16	1234	
16	77	2
16	4	13 = D
	0	4

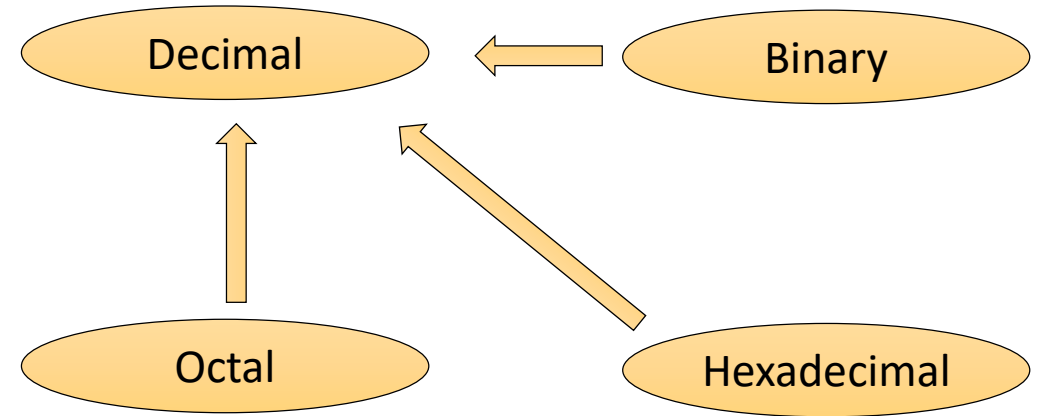


$$(1234)_{10} = (4D2)_{16}$$

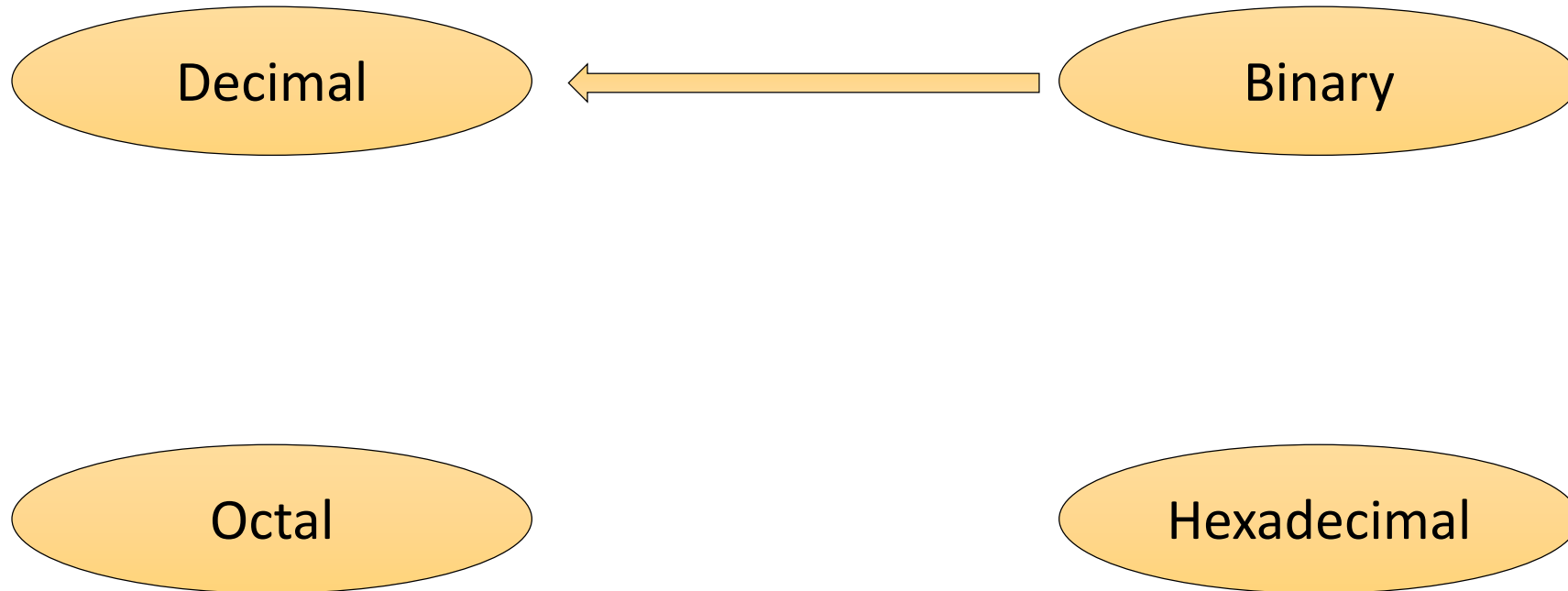
CONVERSION FROM ANY BASE TO DECIMAL

✓ Technique:

1. Multiply each bit by b^n , where n is the “weight” of the bit and b is the base.
2. The weight is the position of the bit, starting from 0 on the right
3. Add the results



CONVERSION FROM BINARY TO DECIMAL



CONVERSION FROM BINARY TO DECIMAL

✓ Example: $(101011)_2 = (N)_{10}$

$$\begin{array}{rcll} 101011_2 & => & 1 & \times 2^0 = 1 \\ & & 1 & \times 2^1 = 2 \\ & & 0 & \times 2^2 = 0 \\ & & 1 & \times 2^3 = 8 \\ & & 0 & \times 2^4 = 0 \\ & & 1 & \times 2^5 = 32 \\ & & & \hline & & & 43_{10} \end{array}$$

$$(101011)_2 = (43)_{10}$$

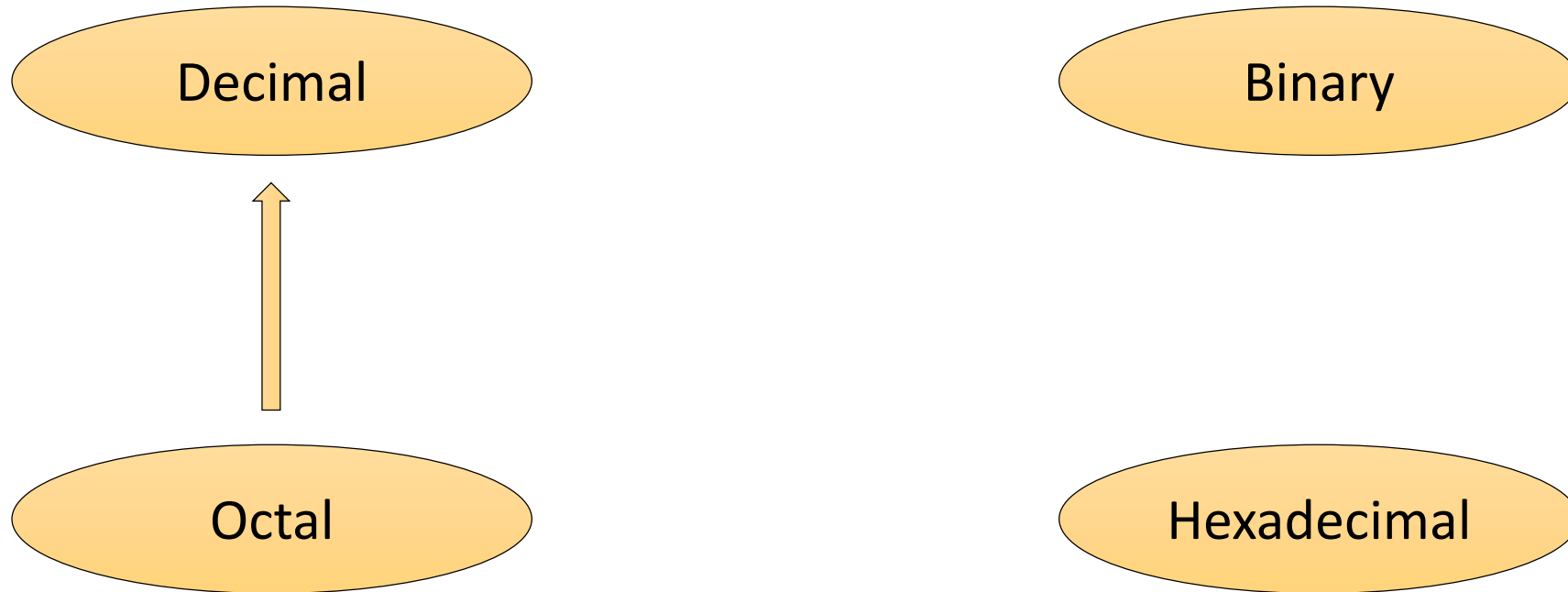
CONVERSION FROM BINARY TO DECIMAL

✓ Example: $(10.1011)_2 = (N)_{10}$

$$\begin{array}{rcll} 10.1011_2 & \Rightarrow & 1 \times 2^{-4} & = 0.0625 \\ & & 1 \times 2^{-3} & = 0.125 \\ & & 0 \times 2^{-2} & = 0.0 \\ & & 1 \times 2^{-1} & = 0.5 \\ & & 0 \times 2^0 & = 0 \\ & & 1 \times 2^1 & = 2 \\ & & \hline & & & 2.6875_{10} \end{array}$$

$$(10.1011)_2 = (2.6875)_{10}$$

CONVERSION FROM OCTAL TO DECIMAL



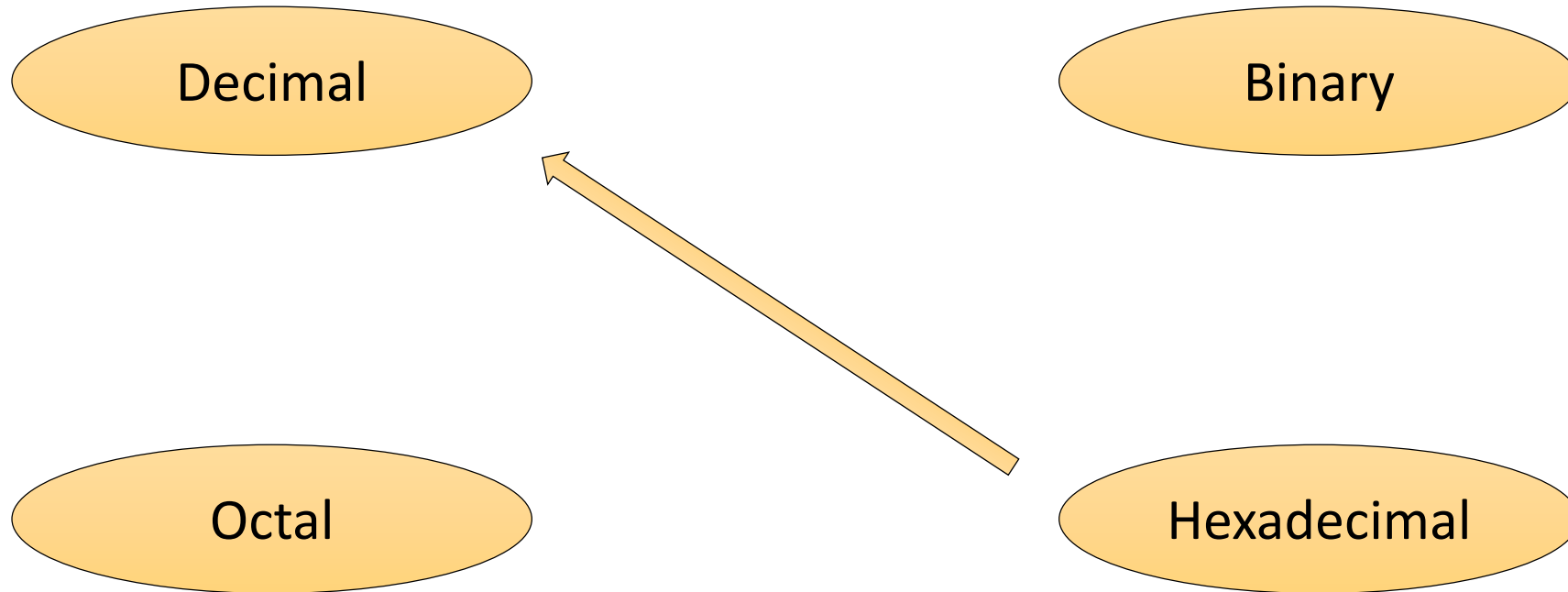
CONVERSION FROM OCTAL TO DECIMAL

✓ **Example:** $(724)_8 = (N)_{10}$

$$\begin{array}{rcll} 724_8 & \Rightarrow & 4 \times 8^0 & = & 4 \\ & & 2 \times 8^1 & = & 16 \\ & & 7 \times 8^2 & = & 448 \\ & & & & \hline & & & & 468_{10} \end{array}$$

$$(724)_8 = (468)_{10}$$

CONVERSION FROM HEXADECIMAL TO DECIMAL



CONVERSION FROM HEXADECIMAL TO DECIMAL

✓ **Example:** $(ABC)_{16} = (N)_{10}$

$$\begin{array}{rcllclcl} ABC_{16} & => & C & \times & 16^0 & = & 12 & \times & 1 & = & 12 \\ & & B & \times & 16^1 & = & 11 & \times & 16 & = & 176 \\ & & A & \times & 16^2 & = & 10 & \times & 256 & = & 2560 \\ & & & & & & & & & & \hline & & & & & & & & & & 2748_{10} \end{array}$$

$$(ABC)_{16} = (2748)_{10}$$

ASSIGNMENT QUESTION

✓ Fill the table:

Decimal	Binary	Octal	Hexa-decimal
33			
	1110101		
		703	
			1AF

Thank You